

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	03 July 2026
Team ID	
Project Name	Cosmetic Insights: Navigating Cosmetics Trends and Consumer Insights with Tableau
Maximum Marks	4 Marks

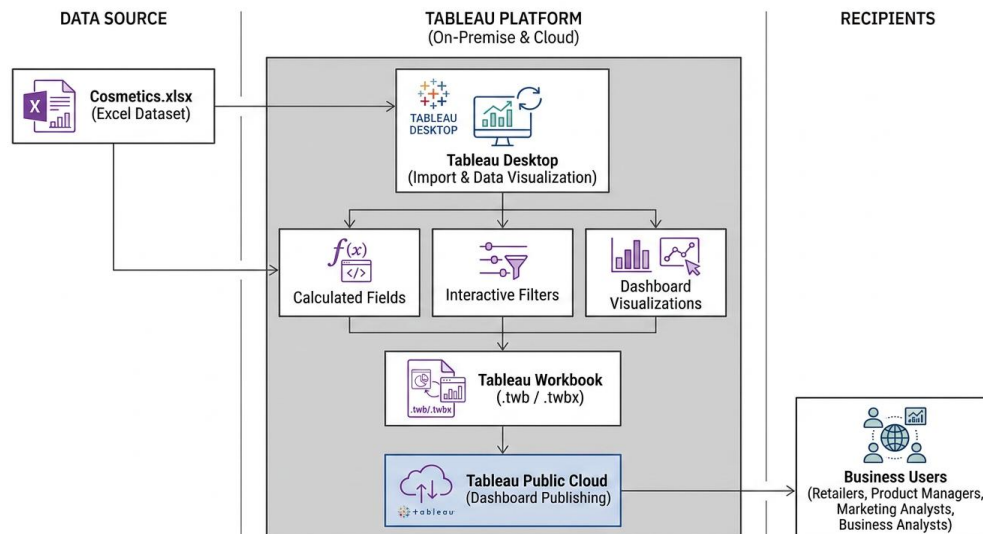
Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>

Technical Architecture Flow Diagram:



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Interactive dashboard used by users to analyze cosmetic product data.	Tableau Dashboard (TableauPublic)
2.	Application Logic	Data import, calculated fields, dashboard creation, interactive filters, and visualizations.	Tableau Public
3.	Database	Cosmetic product data stored in an Excel workbook used for dashboard analysis.	Microsoft Excel (.xlsx)
4.	Cloud Database	Published the interactive dashboard for online access and sharing.	Tableau Public
5.	File Storage	Stores the dataset and Tableau workbook for future analysis and modifications.	Local File System (.xlsx, .twb/.twbx)
6.	Infrastructure (Server / Cloud)	Dashboard developed locally and deployed to the cloud through Tableau Public.	Windows PC + Tableau Public

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	No open-source framework was used in this project. The dashboard was developed using the free version of Tableau Public.	N/A (Tableau Public – Free Edition)
2.	Security Implementations	Dashboard access is managed through the Tableau Public account. The source dataset is stored locally, and only the published dashboard is shared publicly.	Tableau Public Account, Local File System
3.	Scalable Architecture	The dashboard can be extended by adding new datasets, calculated fields, visualizations, filters, and dashboards without redesigning the existing solution.	Tableau Public, Microsoft Excel (.xlsx)

S.No	Characteristics	Description	Technology
4.	Availability	The published dashboard is accessible online through Tableau Public from any device with an internet connection.	Tableau Public Cloud
5.	Performance	Tableau provides optimized rendering of interactive dashboards and quick response to filters and visualizations for the current dataset.	Tableau Public, Microsoft Excel (.xlsx)

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>